

HDOC – 15th day

Name – Piyush Negi

Sap_ID – 590042185

Prepare the algorithm and test case table. Then write, compile, execute, and test the C program against the prepared use cases.

❖ PROBLEM STATEMENT:

Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice:

- Check whether all digits are even.
- Check whether all digits are prime.

○ Algorithm:

- 1) Start.
- 2) Declare variables num, choice, digit, i and an array freq[10].
- 3) Initialize all elements of freq[10] to 0.
- 4) Input a positive integer from the user.
- 5) Display the menu:
 - 1 – Check whether all digits are unique.
 - 2 – Find the frequency of each digit.
- 6) Input the user's choice.
- 7) Use switch(choice) to perform the selected operation.
 - 7.1) If choice is 1:
 - Extract each digit using num % 10.
 - Increase the frequency of that digit.

- Remove the last digit using $\text{num} / 10$.
- Check the frequency of all digits.
- If any digit occurs more than once, display "Not Unique".
- Otherwise, display "Unique".

7.2) If choice is 2:

- Extract each digit using $\text{num} \% 10$.
- Increase its frequency.
- Remove the last digit using $\text{num} / 10$.
- Display the frequency of digits from 0 to 9.

7.3) If the choice is neither 1 nor 2, display "Invalid Choice".

8) Stop.

○ TEST CASE TABLE:

Test case	Input	choice	Expected output	Actual output	Result
1.	248	1	All digits are even	All digits are even	Pass
2.	867	1	All digits are not even	All digits are not even	Pass
3.	37	2	All digits are prime.	All digits are prime.	Pass
4.	21	2	All digits are not prime.	All digits are not prime.	Pass
5.	23	3	Invalid choice	Invalid choice	Pass

○ C PROGRAM:

```
kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1 2 3 4
Open

1#include <stdio.h>
2
3int main()
4{
5    int num, n, digit, choice;
6    int test = 1;
7
8    printf("Enter a positive integer: ");
9    scanf("%d", &num);
10
11    printf("\n1. Check whether all digits are even");
12    printf("\n2. Check whether all digits are prime");
13    printf("\nEnter your choice: ");
14    scanf("%d", &choice);
15
16    n = num;
17
18    switch(choice)
19    {
20        case 1:
21            while(n > 0)
22            {
23                digit = n % 10;
24
25                if(digit % 2 != 0)
26                {
27                    test = 0;
28                    break;
29                }
30
31                n = n / 10;
32            }
33
34            if(test == 1)
35                printf("All digits are even");
36            else
37                printf("All digits are not even");
38
39            break;
40
```

```
kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

1 2 3 4

Open

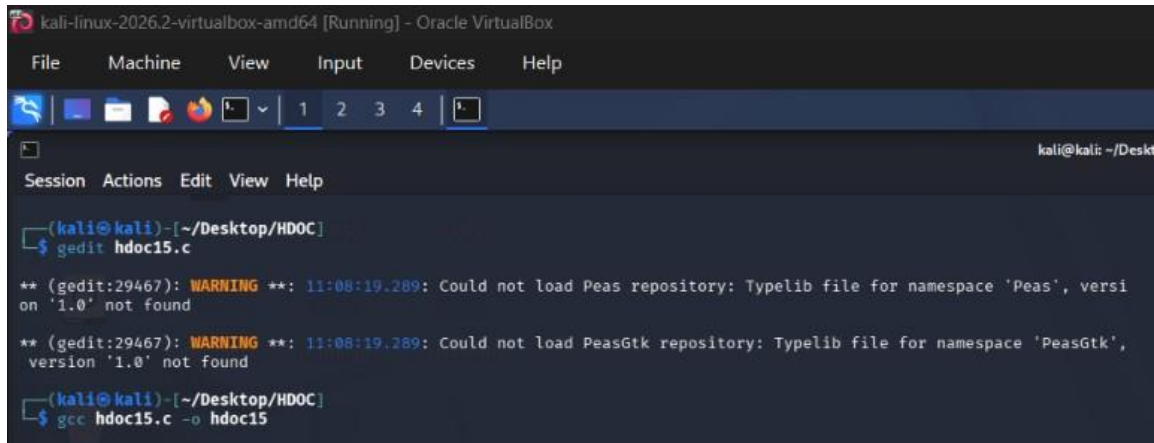
31     n = n / 10;
32     }
33
34     if(test == 1)
35         printf("All digits are even");
36     else
37         printf("All digits are not even");
38
39     break;
40
41
42     case 2:
43         while(n > 0)
44         {
45             digit = n % 10;
46
47             if(digit != 2 && digit != 3 &&
48                digit != 5 && digit != 7)
49             {
50                 test = 0;
51                 break;
52             }
53
54             n = n / 10;
55         }
56
57         if(test == 1)
58             printf("All digits are prime");
59         else
60             printf("All digits are not prime");
61
62         break;
63
64
65     default:
66         printf("Invalid choice");
67     }
68
69     return 0;
70 }
```

Compile the code using:

```
gcc hdoc15.c -o hdoc15
```

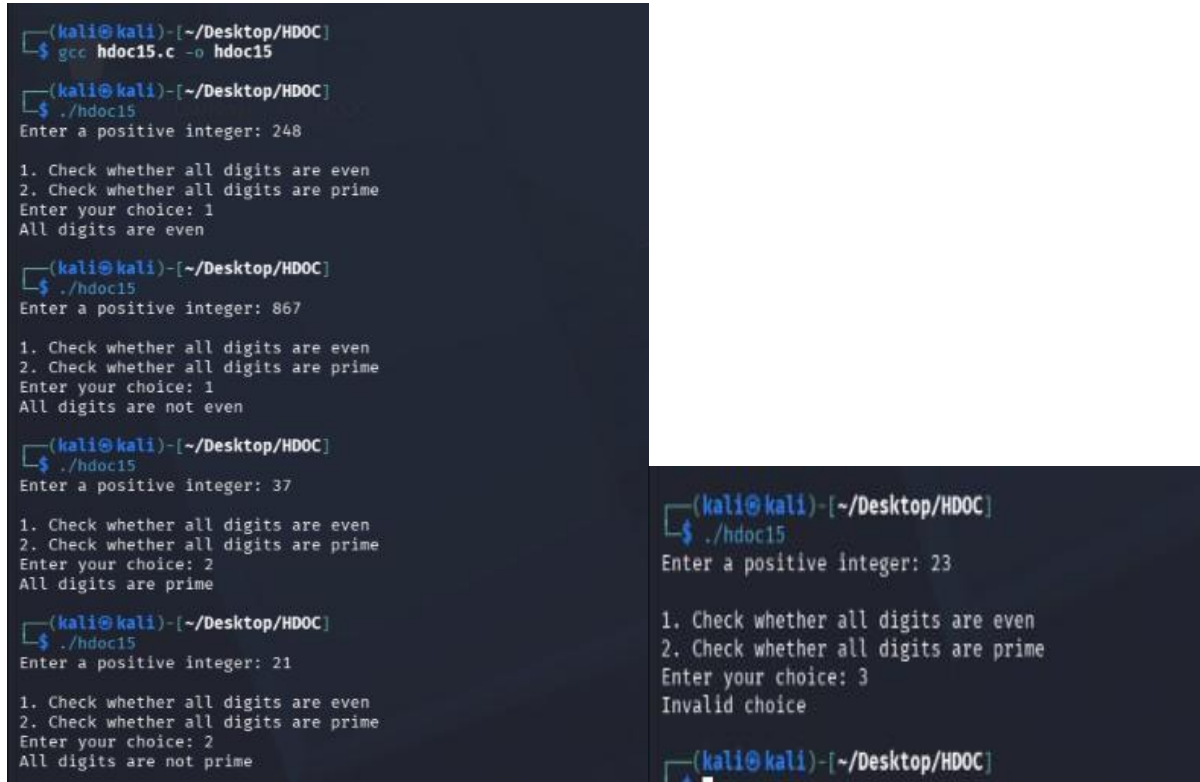
Execute the code using:

```
./hdoc15
```



```
kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1 2 3 4
kali@kali: ~/Desktop
Session Actions Edit View Help
(kali@kali)~[~/Desktop/HDOC]
$ gedit hdoc15.c
** (gedit:29467): WARNING **: 11:08:19.289: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:29467): WARNING **: 11:08:19.289: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
(kali@kali)~[~/Desktop/HDOC]
$ gcc hdoc15.c -o hdoc15
```

Here are some screenshots of the test cases of the code:



```
(kali@kali)~[~/Desktop/HDOC]
$ gcc hdoc15.c -o hdoc15

(kali@kali)~[~/Desktop/HDOC]
$ ./hdoc15
Enter a positive integer: 248

1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 1
All digits are even

(kali@kali)~[~/Desktop/HDOC]
$ ./hdoc15
Enter a positive integer: 867

1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 1
All digits are not even

(kali@kali)~[~/Desktop/HDOC]
$ ./hdoc15
Enter a positive integer: 37

1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 2
All digits are prime

(kali@kali)~[~/Desktop/HDOC]
$ ./hdoc15
Enter a positive integer: 21

1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 2
All digits are not prime

(kali@kali)~[~/Desktop/HDOC]
$ ./hdoc15
Enter a positive integer: 23

1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 3
Invalid choice

(kali@kali)~[~/Desktop/HDOC]
```